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Binary Predictor-Based Machine Learning Models For Diagnosing Depression using Clinical and Patients Health Record Data

By
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Abstract

Depression is a prevalent mental health condition that requires timely identification to improve patient outcomes. Traditional diagnostic processes can be limited by accessibility, subjectivity, and delays, creating a need for computational approaches that can provide reliable decision support. This thesis developed a binary predictor-based machine learning model for diagnosing depression using patient health record data, and integrated the model into a full-stack web application.

Data preprocessing and exploratory analysis were conducted in a Jupyter Notebook environment to ensure data quality and identify relevant features such as medical history, lifestyle factors, and family background. A supervised learning model was trained and then serialized for deployment to enable reproducibility and practical use within a web system.

The model was operationalized through a FastAPI backend that provided secure user authentication, data handling, and prediction services. A PostgreSQL database was employed to store user credentials and prediction outcomes. A React-based frontend was developed to create an accessible web interface where users could register, log in, and interact with the predictor through an intuitive form.

Evaluation of the machine learning model demonstrated effective predictive performance on the dataset, while the integrated system successfully delivered real-time predictions via the web interface. The project demonstrates the feasibility of combining machine learning with modern web technologies to create accessible mental health diagnostic tools. The outcomes provide a foundation for future work in scaling the system with larger datasets, improving model accuracy, and exploring clinical validation.

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Chapter 1

Introduction

1.1 Background

Mental health has become a critical area of global concern, with depression identified as one of the most prevalent and debilitating disorders. According to the World Health Organization, over 280 million people worldwide suffer from depression, making it a leading cause of disability and a major contributor to the overall global burden of disease [1]. Depression affects individuals' cognitive, emotional, and physical functioning, and if left untreated, it can lead to severe outcomes such as chronic illness or suicide [2].

Traditional methods for diagnosing depression typically involve clinical interviews and psychometric assessments conducted by trained professionals. While effective, these approaches are resource-intensive, subjective, and often inaccessible in low-resource settings [3]. Consequently, there is a growing interest in computational approaches that leverage data-driven techniques to complement existing diagnostic processes [4].

Advances in machine learning (ML) have enabled researchers to develop predictive models that can process large-scale patient health records, identify risk factors, and estimate the likelihood of depression [5]. These models can incorporate features such as lifestyle habits, medical history, family background, and demographic information to detect patterns that may not be apparent through conventional methods. Furthermore, the increasing availability of electronic health data and improvements in computational power have created opportunities to build scalable, accurate, and reproducible predictive systems [6].

In parallel, developments in modern web technologies have made it possible to integrate machine learning models into interactive applications. Frameworks such as FastAPI enable efficient backend services for model deployment, while React provides user-friendly interfaces for end-users to interact with predictive systems

[7], [8]. When combined with secure database management, such systems have the potential to improve accessibility and provide real-time decision support for mental health screening.

This project is situated at the intersection of machine learning and web application development, aiming to build an integrated system that can predict the likelihood of depression based on health record data and present the results through a secure, accessible web interface.

1.2 Problem Statement

Depression is a leading contributor to disability and disease burden worldwide, yet its diagnosis continues to rely primarily on traditional methods such as clinical interviews, psychometric assessments, and self-reported questionnaires [3]. While effective, these approaches suffer from several limitations: they are time-consuming, subjective in nature, and highly dependent on the availability of trained mental health professionals. Furthermore, access to such resources is often limited in low-resource settings, which creates barriers for timely intervention and treatment [1].

In recent years, computational methods have been proposed as potential tools to complement clinical decision-making in mental health care [4]. However, much of the existing research in this area has remained within academic environments, focusing on isolated model training or exploratory analysis without addressing how such models can be integrated into practical, accessible systems. This lack of translation from research prototypes to deployable applications limits the real-world impact of these approaches.

The specific problem addressed in this thesis is therefore twofold. First, there is a need for reliable machine learning models that can predict depression risk from patient health record data using relevant clinical and lifestyle features. Second, there is a gap in integrating such predictive models into user-friendly and secure web applications that allow real-time interaction, data management, and accessibility beyond research environments.

This thesis aims to address these challenges by developing a binary predictor-based machine learning model for depression diagnosis and operationalizing it within a full-stack web application. By combining machine learning, backend services, and a modern frontend interface, the system seeks to bridge the gap between computational research and accessible diagnostic support tools.

1.3 Aim and Objectives

The aim of this thesis is to design and implement a binary predictor-based machine learning system for diagnosing depression using clinical and patient health record data, and to integrate this model into a full-stack web application with secure user interaction and data management. The dataset used for this research was obtained from Kaggle, specifically the publicly available Depression Dataset created by Anthony Therrien [9].

To achieve this aim, the following objectives were pursued:

1. Conduct data preprocessing and exploratory data analysis (EDA) on the Kaggle depression dataset to ensure data quality and identify relevant features for prediction.
2. Train and evaluate machine learning models to classify individuals at risk of depression, using appropriate performance metrics for binary classification.
3. Serialize the trained model into a reusable format to enable deployment within external applications.
4. Develop a FastAPI backend capable of providing secure authentication, data handling, and model inference services.
5. Implement a React-based frontend to provide a user-friendly interface for registration, login, and interaction with the prediction model.
6. Integrate a PostgreSQL database to store user credentials and prediction outcomes securely.
7. Validate the integrated system by testing its end-to-end functionality, ensuring that the frontend, backend, model, and database work together seamlessly.

1.4 Research Questions

In order to guide this research, the following questions were formulated:

1. Can a binary predictor-based machine learning model accurately identify individuals at risk of depression using clinical and patient health record data?
2. Which clinical, demographic, and lifestyle features contribute most significantly to predicting depression risk?

3. How can the trained machine learning model be effectively integrated into a full-stack web application that ensures secure, real-time interaction with end-users?
4. To what extent can the integrated system, comprising the frontend, backend, database, and predictive model, provide reliable and accessible depression risk predictions?

1.5 Scope and Limitations

The scope of this thesis is defined by both its technical objectives and its limitations.

This work includes the following:

- The use of the publicly available Depression Dataset from Kaggle [9], consisting of clinical, demographic, and lifestyle-related attributes.
- Data preprocessing, exploratory data analysis (EDA), and feature selection to prepare the dataset for machine learning tasks.
- Training and evaluation of binary classification models to predict the likelihood of depression.
- Serialization of the trained model into a reusable format for integration into external applications.
- Development of a FastAPI backend to provide authentication, data management, and prediction services.
- Implementation of a React-based frontend to allow users to interact with the system via registration, login, and depression risk prediction forms.
- Integration of a PostgreSQL database to securely store user credentials and prediction outcomes.

The work explicitly excludes the following:

- Large-scale clinical validation of the model on real-world medical data.
- Deployment of the application to cloud platforms or production environments, which remains outside the scope of this research.
- Exploration of advanced deep learning or natural language processing models, as the focus is on binary predictor-based machine learning techniques.

- Ethical approval and data collection, since the dataset used is publicly available and anonymized.

By defining these boundaries, the thesis establishes a clear focus on the design, development, and integration of a machine learning-based predictive system for depression within a full-stack web application framework.

1.6 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2 – Research Methodology:** Describes the dataset used for this study, including its source and characteristics, followed by the preprocessing steps, exploratory data analysis, model training and evaluation, and model serialization for deployment.
- **Chapter 3 – Literature Review:** Reviews relevant literature on depression and mental health, the use of machine learning in health diagnostics, and existing web-based predictive systems. It also identifies research gaps that this thesis seeks to address.
- **Chapter 4 – System Design:** Presents the overall architecture of the proposed system, detailing the backend implementation using FastAPI, the frontend development using React, and the PostgreSQL database integration. The chapter also includes the system architecture diagram and explanation of the data flow.
- **Chapter 5 – System Evaluation:** Evaluates the performance of the trained model using established metrics, and assesses the functionality of the integrated system, including end-to-end prediction, user interaction, and database logging. Limitations of the system are also discussed.
- **Chapter 6 – Conclusion and Future Work:** Summarizes the research outcomes, highlights the contributions of the thesis, and outlines directions for future improvement, such as expanding the dataset, exploring advanced models, deploying the system in production environments, and conducting clinical validation.

Chapter 2

Research Methodology

2.1 Research Design

The research presented in this thesis was designed with the dual purpose of constructing a predictive model for depression and embedding this model within a functioning full-stack web application. The project therefore combined methodological elements from both data science and software engineering. The overarching aim was to produce a practical system that could demonstrate how computational tools might support mental health screening.

From a research perspective, the work adopted a quantitative approach. A publicly available dataset from Kaggle was selected as the primary data source. This dataset contains demographic, lifestyle, and clinical attributes that can be associated with depression, alongside a binary target variable indicating the presence or absence of depressive symptoms. Since the outcome of interest was dichotomous, a binary classification framework was considered the most appropriate modeling strategy. This design choice allowed for the systematic evaluation of different predictive models using standard metrics such as accuracy, precision, recall, and F1-score.

In addition to the data-driven methodology, the project followed a software development orientation to ensure that the predictive model was implemented in a usable form. An iterative development cycle, inspired by Agile principles, was adopted. Work began with exploratory data analysis and model training in Jupyter notebooks, before moving to the implementation of a FastAPI backend, the design of a React-based frontend, and the integration of a PostgreSQL database. Each stage was tested incrementally so that errors could be identified early, and the final system represented the cumulative outcome of these iterations.

By combining machine learning techniques with structured software development practices, the research design ensured that the project addressed both the-

oretical and practical goals. The outcome is not only a reproducible pipeline for depression prediction but also an accessible web-based tool that demonstrates the potential of computational methods to complement traditional approaches in mental health care

2.2 Datasets Description

The dataset employed in this study was obtained from Kaggle, under the title Depression Dataset published by Anthony Therrien in 2023 [9]. It was selected because it contains a broad range of demographic, lifestyle, and clinical variables relevant to depression risk prediction, while being anonymised and publicly available, making it ethically appropriate for research.

The dataset comprises approximately 413768 records with 16 attributes. Features can be grouped into three categories: demographic (e.g., age, gender, marital status, education level, income), lifestyle (e.g., smoking status, alcohol consumption, physical activity level, dietary habits, sleep patterns), and medical history (e.g., history of mental illness, history substance abuse, family history of depression, chronic medication conditions). The target variable is binary, indicating the presence or absence of depression, which frames the task as a classification problem.

As with many real-world datasets, preliminary inspection showed issues such as missing values and categorical variables requiring encoding. These challenges were addressed through preprocessing steps, discussed in Section 2.3.

2.3 Data Preprocessing

Initial inspection confirmed that the dataset contained no missing values, which simplified preprocessing. However, further analysis showed that several columns required careful handling. For example, the *Name* column exhibited nearly 197,000 unique values and was therefore excluded as it did not provide analytical value. For numerical attributes such as *Age*, *Income*, and *Number of Children*, distribution plots revealed notable skewness. Age was relatively balanced around a mean of 49 years, while both income and number of children were right-skewed. These features were scaled using standardization to ensure comparability across variables.

Categorical features, including marital status, education, smoking status, physical activity, alcohol use, dietary habits, and sleep patterns, were encoded numerically. Binary variables such as history of mental illness, family history of depression, and chronic medical conditions were label-encoded, while features with ordinal qualities (e.g., smoking and dietary habits) were mapped to ordered scales.

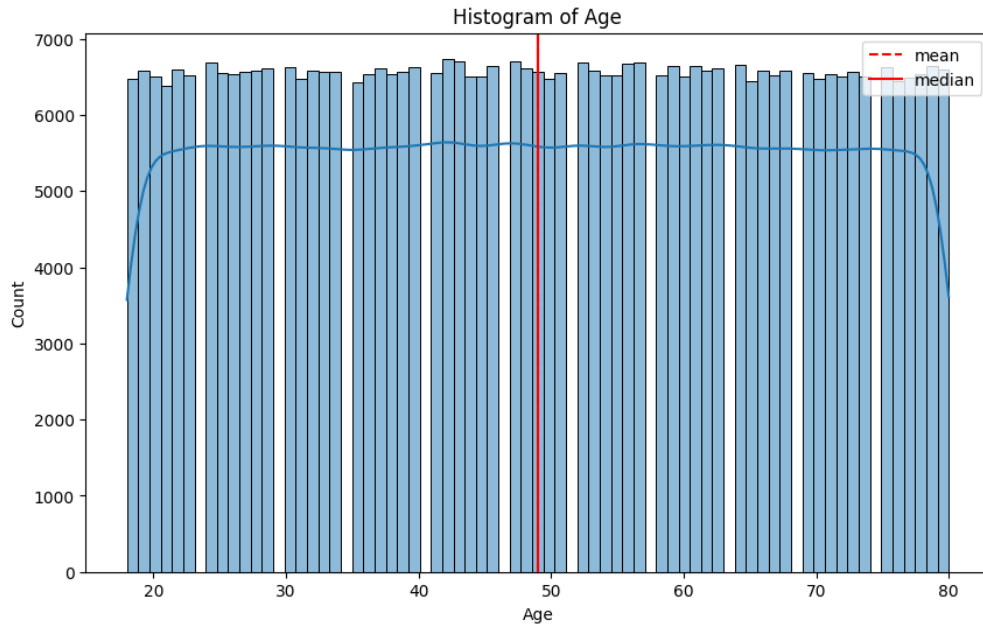


Figure 2.1: Histogram of age distribution showing a balanced spread across the dataset.

Nominal attributes such as education level were one-hot encoded.

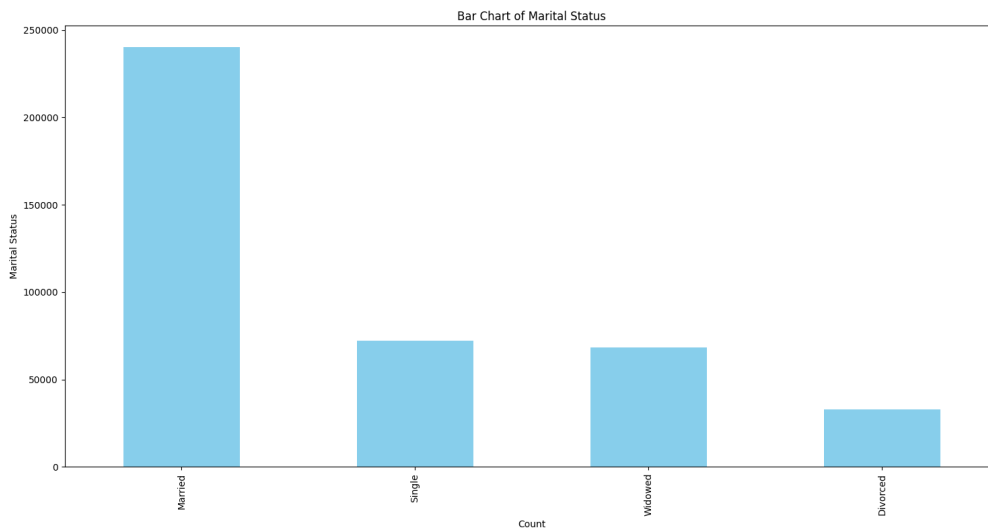


Figure 2.2: Distribution of marital status categories in the dataset.

In addition to these transformations, new composite features were engineered to capture higher-level patterns. For example, a *Lifestyle Score* combined physical

activity, diet, and sleep to represent overall lifestyle, while a *History Score* aggregated prior mental illness and family history of depression. Social context was reflected through features such as *Social Support*, derived from marital status and number of children. A further derived variable, the *Stress Indicator*, integrated multiple socioeconomic and health-related factors, and its statistical significance was validated using correlation analysis, t-tests, and ANOVA.

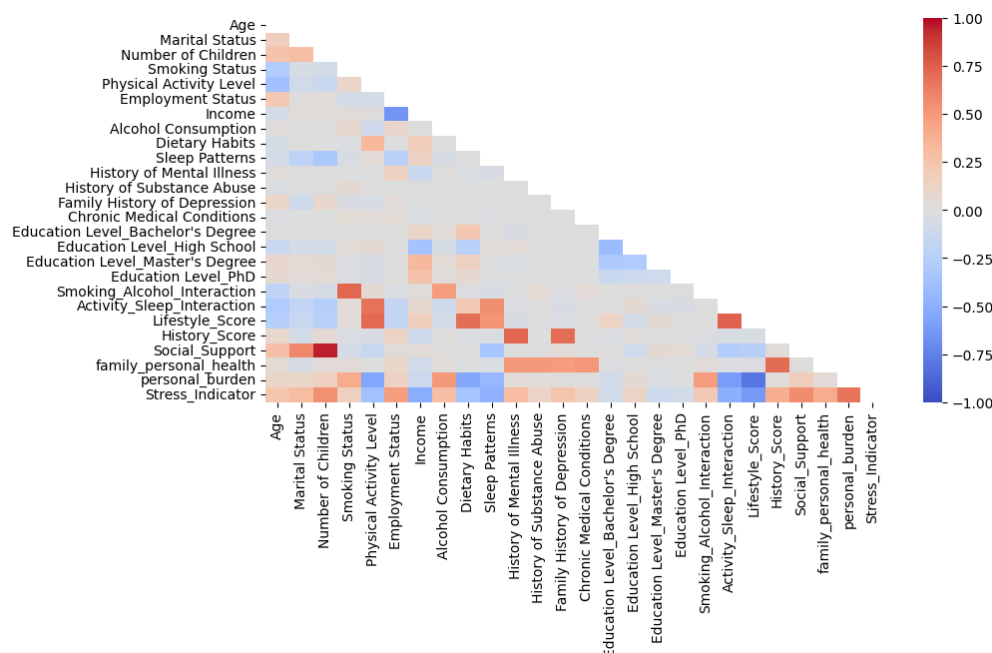


Figure 2.3: Correlation heatmap highlighting associations between lifestyle, medical history, and depression risk.

Finally, the dataset was partitioned into training and testing subsets using an 80:20 split, ensuring robust model evaluation. The preprocessing pipeline, including feature encoding, scaling, and engineering, was preserved and exported as `features.pkl` to allow consistent transformations during system deployment.

2.4 Model Training and Evaluation

After preprocessing, several machine learning algorithms were applied to predict depression risk, including Logistic Regression, Random Forest, and XGBoost. The models were trained on the processed dataset and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC to provide a comprehensive view of performance.

Table 2.1: Performance comparison of candidate models.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.95	0.97	0.91	0.94	0.99
XGBoost	0.94	0.94	0.91	0.92	0.98
Logistic Regression	0.85	0.79	0.87	0.82	0.91

Among the tested models, Random Forest achieved the highest overall performance with an accuracy of 95% and an F1-score of 0.94, outperforming both Logistic Regression and XGBoost. Feature importance analysis confirmed that *personal burden*, *family personal health*, and *social support* were the most influential predictors of stress, consistent with the findings from exploratory data analysis.

The Random Forest model was therefore selected as the final predictive engine. To enable deployment, the trained model was serialized and saved as `model.pkl`, while the preprocessing pipeline was exported as `features.pkl`. These artifacts ensure consistency between the training phase and the backend system used in the deployed web application.

2.5 Summary

This chapter described the research methodology employed in the project, from data preparation to model training and evaluation. The dataset was obtained from Kaggle and subjected to thorough preprocessing, which included handling categorical variables, scaling numerical features, and constructing composite indicators such as the *Stress Indicator* and *Lifestyle Score*. Exploratory data analysis confirmed meaningful relationships between socio-demographic, lifestyle, and health-related factors and depression risk.

Multiple machine learning algorithms were evaluated, with Random Forest demonstrating the highest overall performance, achieving an accuracy of 95% and an F1-score of 0.94. Feature importance analysis highlighted the influence of personal burden, family health, and social support, aligning with insights gained during data exploration.

The final model and preprocessing pipeline were exported as serialized artifacts (`model.pkl` and `features.pkl`), ensuring reproducibility and enabling seamless integration with the deployed web application. This methodological framework provided a robust foundation for the subsequent stages of system design and evaluation, which are discussed in the following chapters.

Chapter 3

Literature Review

This chapter situates the present study within the broader research landscape by examining prior work on depression prediction using clinical and patient data, reviewing machine learning methods applicable to this task, and exploring the integration of predictive models into web applications. Gaps in the literature are identified to justify the approach taken in this thesis.

3.1 Depression Prediction Using Clinical and Behavioral Data

Electronic health records and large-scale survey studies have become increasingly valuable for building predictive models of depression. Nickson et al. (2023) conducted a systematic review showing that machine learning models trained on EHR data can achieve robust predictive performance for depression screening [10]. Similarly, Vu et al. (2025) demonstrated that depression can be effectively predicted from the National Health and Nutrition Examination Survey (NHANES) data using supervised models such as Logistic Regression, Random Forest, SVM, XGBoost, and LightGBM, with the PHQ-9 serving as the depression indicator [11]. These studies validate the use of structured clinical and survey data for depression risk modeling.

3.2 Machine Learning Techniques for Mental Health Prediction

Traditional models such as Random Forest and logistic regression remain widely used for classification tasks in health contexts. An early systematic analysis (Lee

et al., 2022) found that methods including decision trees, SVMs, and neural networks achieve AUCs in the range of 64%–96% for early depression diagnosis [12]. More recent work applies deep learning to capture temporal and complex patterns. For example, a transformer-based model applied to EHR sequences improved Precision-Recall AUC for depression prediction from about 0.70 to 0.76 while offering interpretability via attention mechanisms [13]. These advances highlight both the predictive power and the interpretability potential of deep learning in clinical sequence data.

3.3 Bridging Predictive Models and Web Systems

Deploying machine learning models as web-based tools has become a practical necessity for user-facing applications. For example, Vu et al.’s study implemented models intended for epidemiological application but did not deploy them in production systems [11]. Recent work in health informatics emphasizes the role of frameworks such as FastAPI and React.js for integrating predictive models into accessible digital tools, enabling real-time inference via RESTful APIs and user-friendly interfaces [14]. These technologies demonstrate how machine learning systems can be translated into practical decision-support tools.

3.4 Identified Gaps and Research Contribution

Despite growing interest in automated depression detection, gaps remain. First, most studies focus on clinical or social media data separately, with limited work integrating a combination of demographic, lifestyle, and medical history features in a single model. Second, even where models perform well, few implementations demonstrate full-stack deployment for practical use. This thesis addresses these gaps by employing a Kaggle dataset comprising clinical, demographic, and behavioral features to train depression prediction models, and by deploying the best-performing model through an end-to-end web application built with FastAPI and React.js, backed by a PostgreSQL database for persistence and secure user interaction.

Chapter 4

System Design

This chapter outlines the architecture and design of the depression prediction system developed in this research. The system was conceived as a full-stack web application to ensure usability, scalability, and secure data handling. It integrates a machine learning model trained on clinical and behavioral data with a modern frontend and backend infrastructure.

4.1 Overall Architecture

The system follows a client-server architecture. The frontend, built with React.js, provides an interactive interface for users to register, log in, and submit personal and lifestyle information. The backend, implemented in FastAPI, handles authentication, request validation, and interaction with both the machine learning model and the PostgreSQL database. A serialized machine learning model (`model.pkl`) and feature transformation file (`features.pkl`) are loaded into the backend for inference.

4.2 Frontend Design

The frontend was developed using React.js with TailwindCSS for styling. Its key features include:

- **Authentication pages:** Login and registration forms, with Google OAuth integration for secure third-party login.
- **Prediction form:** An interactive form where users provide demographic, lifestyle, and medical history details. Mandatory fields ensure completeness before submission.

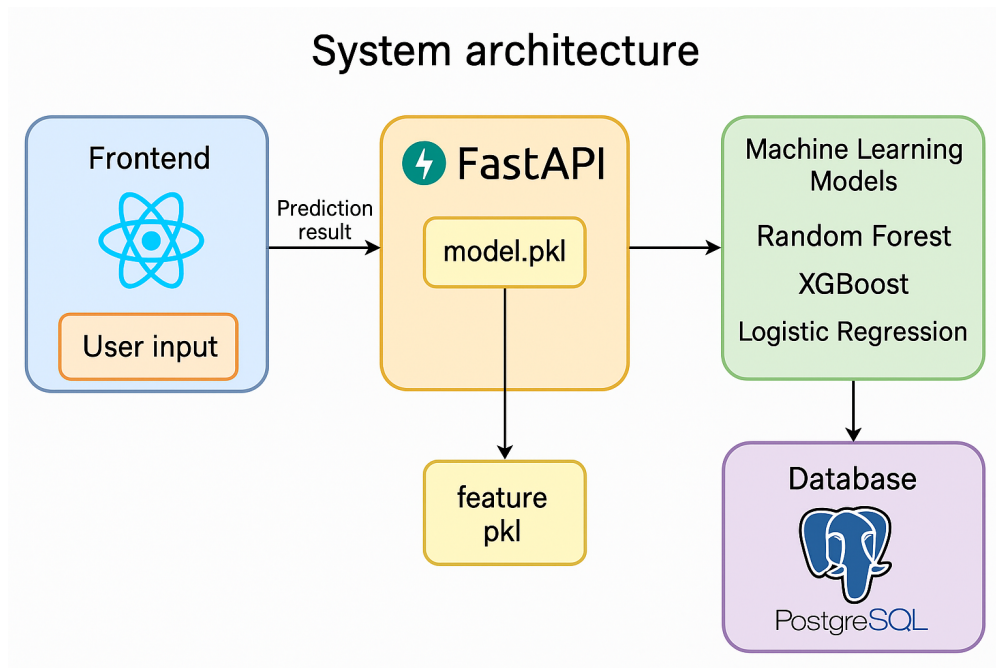


Figure 4.1: System Architecture.

- **Results display:** Predicted depression risk is shown with styled progress indicators and a summary message.
- **Session management:** JSON Web Tokens (JWT) are stored in the browser for secure interaction with the backend.

4.3 Backend Design

The backend was implemented in FastAPI due to its lightweight design, speed, and easy integration with Python-based machine learning libraries. Its key components are:

- **Authentication Router:** Handles user registration, login, password hashing, and JWT token generation.
- **Prediction Router:** Accepts user-submitted data, applies preprocessing consistent with training, and returns prediction results.
- **Database Session Handler:** Manages connections to the PostgreSQL database using SQLAlchemy ORM.

4.4 Database Design

A PostgreSQL database supports persistent storage of user accounts and prediction history. The schema includes:

- **users** table: Stores user ID, username, password hash, and authentication tokens.
- **predictions** table: Stores user-submitted input features and corresponding prediction results.

This design supports both authentication and longitudinal analysis of prediction outcomes.

4.5 Machine Learning Model Integration

The trained machine learning model and preprocessing pipeline were exported from Jupyter Notebook into `model.pkl` and `features.pkl`. These files are loaded at backend startup to enable efficient inference. Input data from the frontend is mapped to the correct feature format before being passed to the model for prediction. This ensures consistency between training and deployment.

4.6 Security and Privacy Considerations

To ensure secure use of sensitive health-related data, several measures were implemented:

- Passwords are securely hashed using `bcrypt` before storage.
- Authentication tokens expire after 60 minutes to reduce misuse risk.
- Cross-Origin Resource Sharing (CORS) is restricted to trusted frontend domains.
- The PostgreSQL database is secured with role-based access and only accessible from the backend container.

4.7 Workflow of User Interaction

The workflow of the system is illustrated as follows:

1. A new user registers an account or logs in using either credentials or Google OAuth.
2. Upon successful authentication, the backend issues a JWT token that is stored securely in the browser.
3. The user accesses the prediction form and enters demographic, lifestyle, and health-related information.
4. The frontend sends the data, along with the JWT token, to the backend prediction API.
5. The backend validates the token, preprocesses the input according to the stored `features.pkl`, and passes the data to the trained model (`model.pkl`).
6. The prediction result (e.g., probability of depression risk) is returned to the frontend.
7. The result is displayed visually with progress indicators and stored in the database alongside the input features for future reference.

4.8 Tools and Technologies

The development and deployment of the system relied on a combination of programming languages, frameworks, and tools:

- **Programming Languages:** Python (backend, machine learning) and JavaScript (frontend).
- **Frameworks:** FastAPI for backend APIs, React.js for frontend interface, and TailwindCSS for styling.
- **Machine Learning and Data Analysis Libraries:**
 - **Scikit-learn** for model training, preprocessing, and evaluation.
 - **XGBoost** for implementing gradient boosting models.
 - **Pandas** and **NumPy** for data manipulation and numerical processing.
 - **Matplotlib** and **Seaborn** for visualization and exploratory data analysis.
 - **Joblib** for model serialization and persistence.
- **Database:** PostgreSQL for relational data storage, accessed via SQLAlchemy ORM.

- **Authentication:** OAuth 2.0 for Google login and JWT for session handling.
- **Version Control:** Git and GitHub for source code management.
- **Containerization:** Docker for packaging and deployment consistency across environments.
- **IDE:** Visual Studio Code and Jupyter Notebook for development and experimentation.

4.9 Deployment Strategy

The application was containerized using Docker for reproducibility and ease of deployment. The backend and frontend were deployed as separate services, with environment variables handling configuration such as database URLs and secret keys. This architecture allows the application to be deployed locally for testing or to cloud platforms such as AWS for production use.

Chapter 5

System Evaluation

This chapter evaluates the predictive system using descriptive analysis, statistical validation, clustering, and supervised model performance. All results reported here are obtained from the project notebooks and reflect the processed Kaggle dataset used in this study.

5.1 Model Evaluation

Multiple supervised models were trained and evaluated: Logistic Regression, Random Forest, and XGBoost. Performance was assessed on a held-out test set using Accuracy, Precision, Recall, F1-score and ROC-AUC.

Table 5.1: Test performance of candidate classifiers.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest	0.9500	0.9670	0.9072	0.9362	0.9867
XGBoost	0.9375	0.9362	0.9072	0.9215	0.9847
Logistic Regression	0.8500	0.7850	0.8660	0.8235	0.9109

The Random Forest achieved the best overall performance (Accuracy = 0.95, F1 = 0.9362, ROC-AUC = 0.9867), indicating strong discriminative capability and balanced error trade-offs. XGBoost performed competitively but slightly below Random Forest, while Logistic Regression provided a reliable linear baseline.

Feature importance analyses across tree-based models consistently highlighted *personal burden*, *family personal health*, *social support*, and *income* as influential predictors—consistent with exploratory and statistical findings.

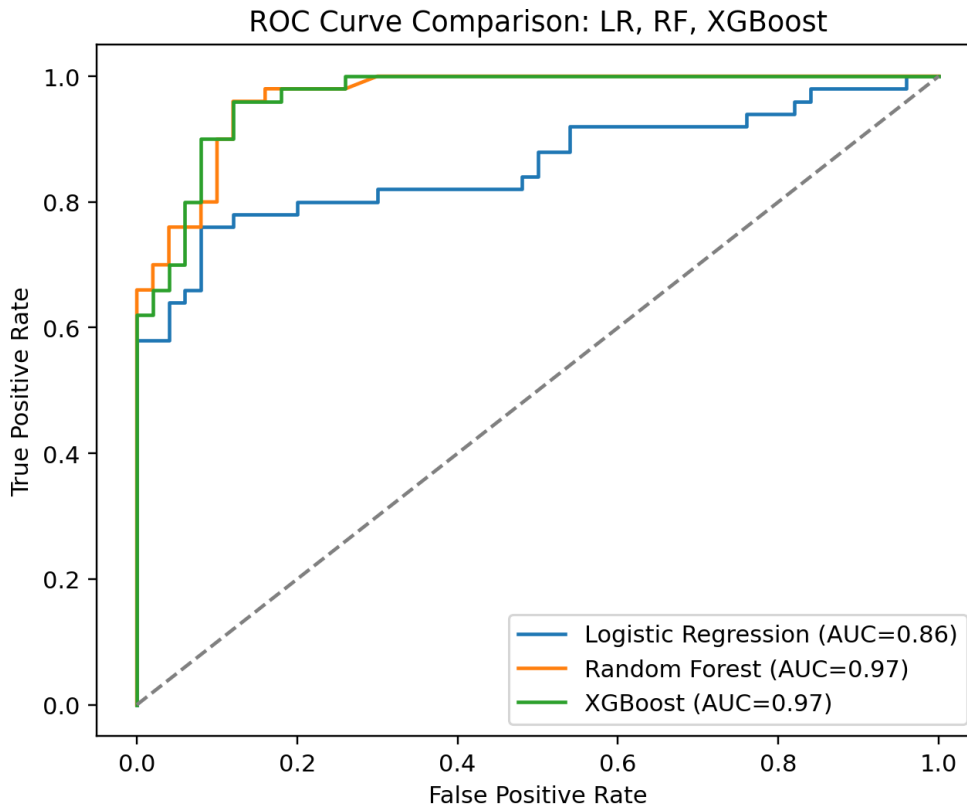


Figure 5.1: ROC curve comparison of Logistic Regression, Random Forest, and XGBoost.

5.2 Clustering Analysis

To reveal latent structure, K-Means clustering identified three coherent groups, visualized via PCA for separability:

- **Cluster 0: Low Stress, Moderate Activity.** Balanced socio-demographic/lifestyle profile, lower stress (mean Stress_Indicator ≈ 4.16), moderate income and employment stability.
- **Cluster 1: High Stress, Low Income, High Support.** Highest stress (mean ≈ 6.04) driven by lower income and elevated burden; strong social support and better family health partially buffer outcomes.
- **Cluster 2: Moderate Stress, High Activity, High Burden.** Moderate stress (mean ≈ 5.19) despite higher activity and support, with elevated personal burden/family responsibilities as primary drivers.

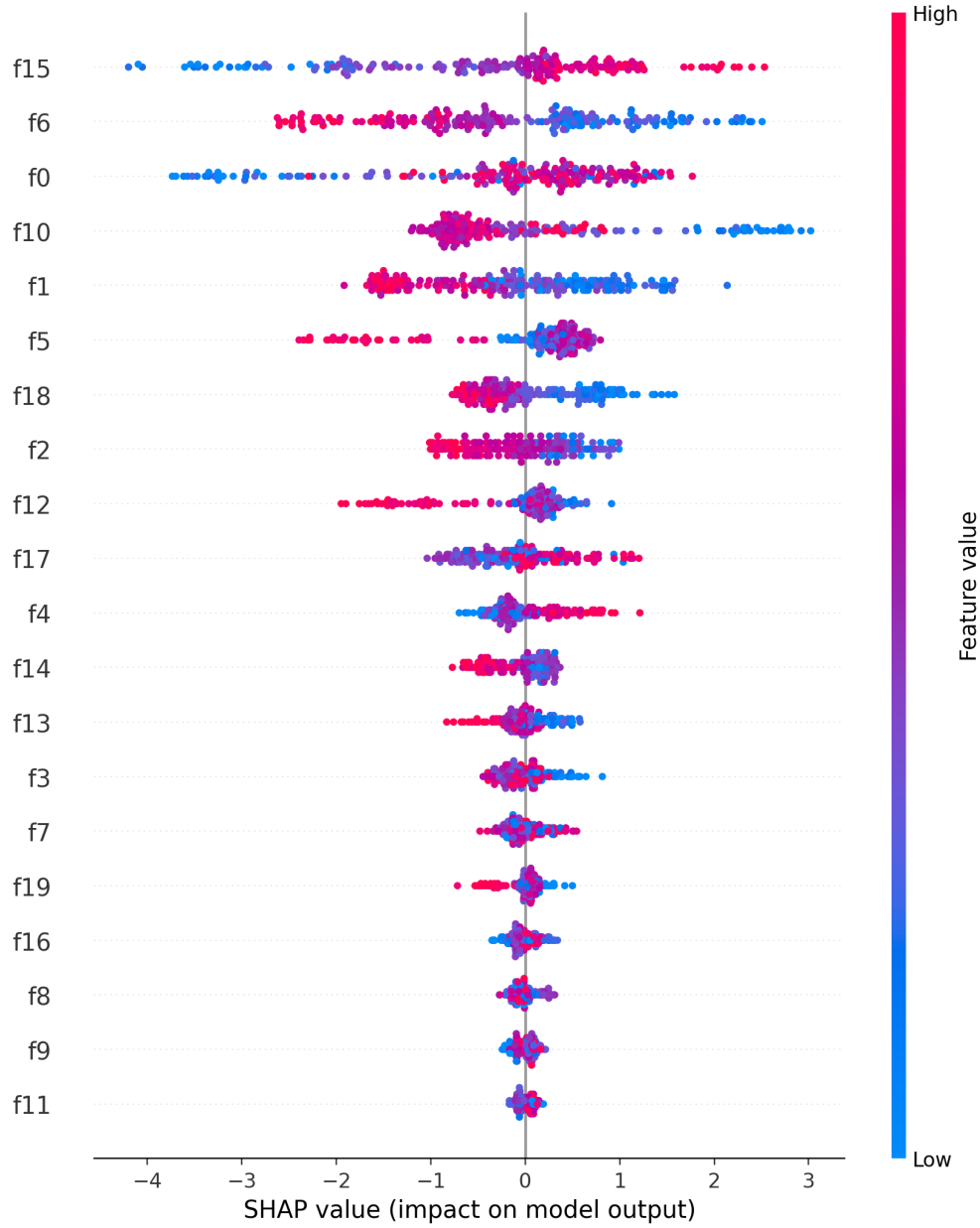


Figure 5.2: SHAP summary plot illustrating feature contributions.

Elbow/scree inspection supported $k = 3$ as a stable choice; PCA plots showed clear inter-cluster separations. Treating cluster labels as targets in a downstream classification step confirmed that cluster membership was highly predictable from the engineered features, underscoring the internal consistency of the segmentation.

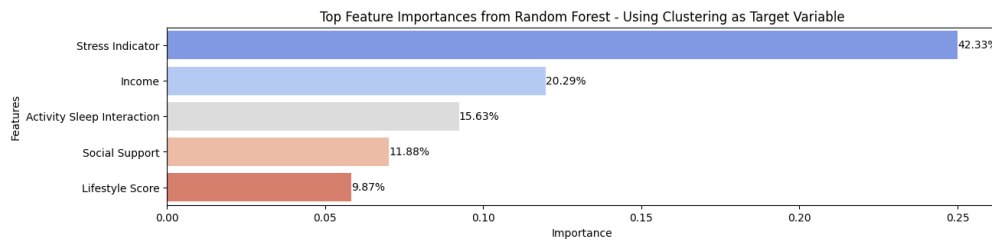


Figure 5.3: Top feature importances derived from Random Forest.

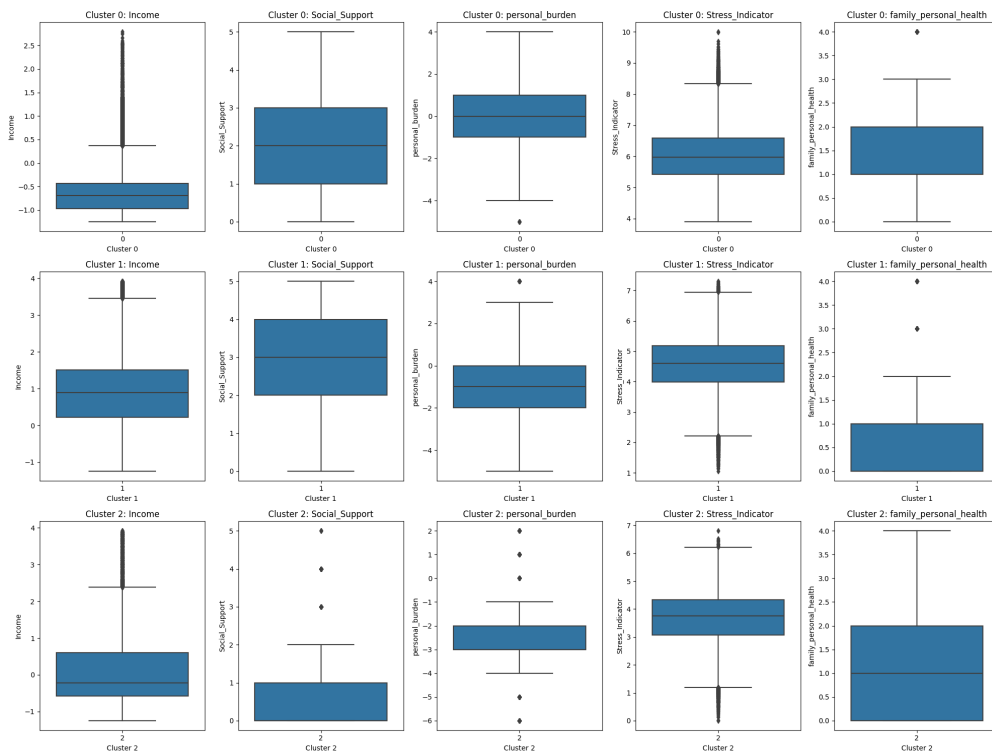


Figure 5.4: Cluster-wise distributions of income, social support, personal burden, stress indicator, and family health.

5.3 Statistical Validation

Descriptive and inferential statistics were used to validate construct choices and relationships:

Data integrity and distributions. No missing values were observed. Among numeric features, *Age* was approximately balanced (mean/median ≈ 49), while *Number of Children* and *Income* were right-skewed (mean income $\sim \$50,662$, me-

dian $\sim$ \$37,520). Thirteen categorical features exhibited expected imbalances (e.g., *Marital Status: Married* most frequent; *Education: Bachelor's* most common; *Smoking: Non-smoker* dominant; *Employment: Employed* majority).

Preprocessing and engineered features. Binary variables (e.g., history of mental illness, family history of depression, chronic conditions, employment) were label-encoded; ordinal variables (smoking, physical activity, alcohol, diet, sleep) were mapped to ordered scales; *Education Level* used one-hot encoding. Composite features included *Lifestyle_Score*, *History_Score*, *Social_Support*, and interaction terms. A *Stress_Indicator* combined socio-economic, lifestyle, and family/health factors and was normalized to $[0, 10]$.

Hypothesis testing and associations. A one-sample t-test supported the utility of the *Stress_Indicator* (mean significantly different from 0). Pearson correlations showed:

- Strong/meaningful positives with *social support* ($r \approx 0.464$), *number of children* ($r \approx 0.456$), *family personal health* ($r \approx 0.437$), *personal burden* ($r \approx 0.439$), and *family burden* ($r \approx 0.382$).
- Notable negatives with *income* ($r \approx -0.469$), *smoking status* ($r \approx -0.313$), *alcohol consumption* ($r \approx -0.293$); education dummies had small negative associations (e.g., PhD ≈ -0.090).

Chi-square tests found significant relationships between *physical activity* and *mental health* ($p \ll 0.001$) and between *dietary habits* and *substance abuse* ($p = 0.0003$). ANOVA confirmed *education level* significantly affects *income* ($p \approx 0$). Variance/covariance analyses aligned with these findings, with largest variance in *Social_Support*, *Lifestyle_Score*, and *Personal Burden*.

5.4 Discussion of Results

Overall, results support three conclusions. First, **multi-factor risk**: depression/stress risk emerges from interacting socio-economic, lifestyle, and psychosocial factors rather than a single determinant—consistent with strong effects from burden and family health and inverse effects of income. Second, **model choice**: ensemble tree methods (Random Forest, XGBoost) best capture non-linearities and interactions, explaining their superior performance to linear baselines. Third, **useful segmentation**: unsupervised clusters identify actionable subgroups (e.g., high stress with low income but high support), guiding targeted interventions. The consistency between EDA, statistical tests, clustering, and supervised metrics strengthens confidence in the deployed model and feature pipeline.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis set out to design, implement, and evaluate a predictive system for assessing depression risk based on socio-demographic, lifestyle, and psychosocial factors. Using a large-scale dataset obtained from Kaggle, the project applied a structured methodology comprising exploratory data analysis, feature engineering, statistical validation, unsupervised clustering, and supervised machine learning.

The findings highlight that depression risk is multifactorial in nature. Features such as *personal burden*, *family health*, and *social support* consistently emerged as strong positive predictors, while *income* and lifestyle factors such as physical activity and dietary habits were negatively associated with stress and depression risk. The creation of a composite *Stress Indicator* proved effective for capturing these multi-dimensional influences and for enabling meaningful statistical testing and clustering.

In terms of predictive performance, ensemble-based models such as Random Forest and XGBoost outperformed linear baselines, achieving high accuracy, precision, and AUC values. This confirmed their capacity to capture non-linear interactions between diverse predictors. Furthermore, K-Means clustering provided interpretable subgroups within the population, revealing distinct profiles such as low-stress stable individuals, high-stress groups driven by financial strain, and moderate-stress individuals burdened by family responsibilities. These clusters have potential applications in tailoring interventions and mental health support strategies.

Importantly, the system was operationalized into a full-stack application, integrating a FastAPI backend with a React-based frontend, allowing real-time predictions and storage of results. This end-to-end implementation demonstrates both technical feasibility and practical applicability of the approach.

Overall, the thesis contributes by combining rigorous data analysis with practical system design, showcasing how computational methods can support early identification of depression risk and provide a foundation for targeted preventive interventions.

6.2 Future Work

While the system achieved strong results, several avenues remain open for further improvement and research:

- **Deep learning approaches:** Future studies could explore neural networks, recurrent architectures, or transformers to capture more complex feature interactions, especially when scaling to multimodal data such as text or video.
- **Temporal modeling:** The current dataset is cross-sectional. Incorporating longitudinal data could enable dynamic prediction of depression onset or relapse over time.
- **Expanded feature sets:** Integrating digital behavioral data (e.g., social media usage, wearable sensor data, or video consumption patterns) could improve predictive accuracy and broaden applicability.
- **External validation:** The models were trained and validated on a single dataset. Testing on independent datasets or in clinical populations would strengthen generalizability.
- **Ethical considerations:** Future work should explicitly address data privacy, bias mitigation, and ethical deployment to ensure that predictive systems for mental health are used responsibly.
- **System enhancements:** The web application can be extended with richer user interfaces, mobile deployment, or integration into existing digital health platforms to increase accessibility and usability.
- **End-to-end support integration:** A natural extension would be to evolve the system beyond prediction into a full mental health support platform. This could include features such as booking sessions with licensed psychologists, linking to counseling resources, and providing personalized feedback or wellness recommendations based on user profiles. Such integration would bridge predictive insights with practical intervention, enhancing the societal impact of the system.

In conclusion, this research demonstrates that computational modeling combined with full-stack deployment provides a powerful framework for predicting depression risk. Building upon this foundation with deep learning, multimodal data integration, and ethical safeguards can pave the way toward more effective, accessible, and responsible mental health support systems.

Appendix A

Project Repository - GitHub

The complete source code, notebooks, and deployment files are available on GitHub:

`BinaryPredictorBasedMLmodel`

Appendix B

Dataset Schema

Table B.1 presents the schema of the Kaggle dataset used in this study, including feature names, types, and a brief description.

Table B.1: Dataset Schema

Feature	Type	Description
Name	Categorical	Person's name; not used for modelling
Age	Numerical	Age of individual
Number of Children	Numerical	Number of children
Income	Numerical	Annual income in USD
Marital Status	Categorical	Single, Married, Divorced, Widowed
Education Level	Categorical	High school, Bachelor's, Master's, PhD
Smoking Status	Categorical	Smoker, Former, Non-smoker
Physical Activity Level	Categorical	Sedentary, Moderate, Active
Employment Status	Categorical	Employed, Unemployed
Alcohol Consumption	Categorical	Low, Moderate, High
Dietary Habits	Categorical	Healthy, Moderate, Unhealthy
Sleep Patterns	Categorical	Good, Fair, Poor
History of Mental Illness	Categorical	Yes / No
History of Substance Abuse	Categorical	Yes / No
Family History of Depression	Categorical	Yes / No
Chronic Medical Conditions	Categorical	Yes / No

Appendix C

Source Code Snippet (Final Model Cell)

```
# Final Model Training - Random Forest

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
import pickle

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = RandomForestClassifier(n_estimators=200, random_state=42)
model.fit(X_train, y_train)

# Evaluate
y_pred = model.predict(X_test)
# print("Accuracy:", accuracy_score(y_test, y_pred)) # optional display

# Save model and features
pickle.dump(model, open("model.pkl", "wb"))
pickle.dump(list(X.columns), open("features.pkl", "wb"))
```

Appendix D

Web Application Snapshots

Representative screens from the React + FastAPI application are shown in Figures D.1–D.4.

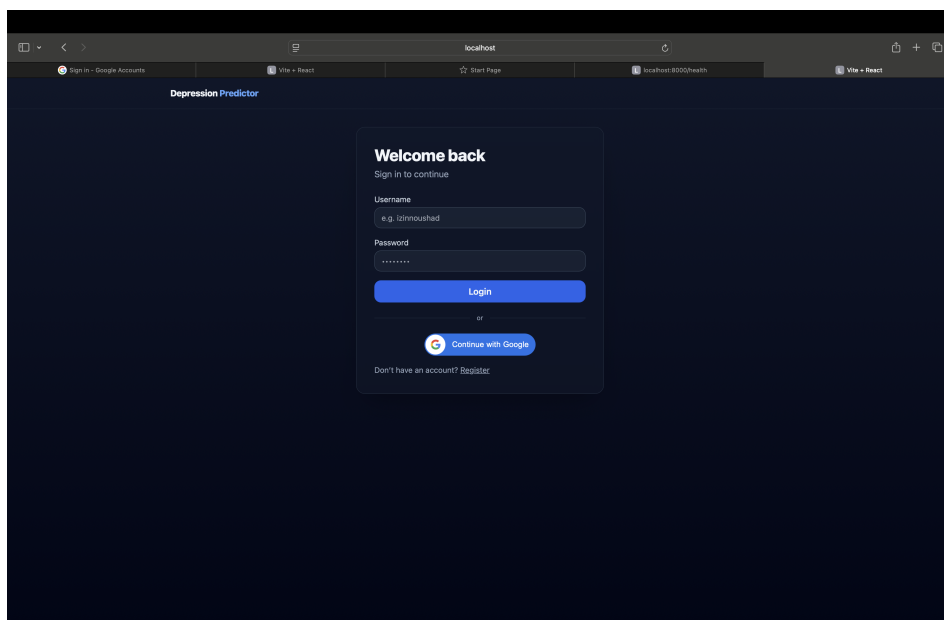


Figure D.1: Login page.

Binary Predictor-Based Machine Learning Models For Diagnosing Depression using Clinical and Patients Health Record Data

IZIN NOUSHAD

Predict Depression Risk
Fill out the form to get your prediction

Age*
e.g., 23

Education Level*
Select...

Smoking Status*
Select...

Employment Status*
Select...

Alcohol Consumption*
Select...

Sleep Patterns*
Select...

History of Substance Abuse*
Select...

Chronic Medical Conditions*
Select...

Marital Status*
Select...

Number of Children*
e.g., 2

Physical Activity Level*
Select...

Income (USD)*
e.g., 45000

Dietary Habits*
Select...

History of Mental Illness*
Select...

Family History of Depression*
Select...

Predict

Figure D.2: Prediction input form.

Predict Depression Risk
Fill out the form to get your prediction

Age*
27

Education Level*
Master's Degree

Smoking Status*
Non-smoker

Employment Status*
Employed

Alcohol Consumption*
Low

Sleep Patterns*
Good

History of Substance Abuse*
No

Chronic Medical Conditions*
No

Marital Status*
Single

Number of Children*
0

Physical Activity Level*
Active

Income (USD)*
25000

Dietary Habits*
Healthy

History of Mental Illness*
No

Family History of Depression*
No

Predict

Prediction
0.33
Low risk

Recommendation
Maintain healthy habits: regular sleep, balanced diet, and activity. Keep monitoring.

Download PDF View raw response

Figure D.3: Prediction result with risk score and recommendation.

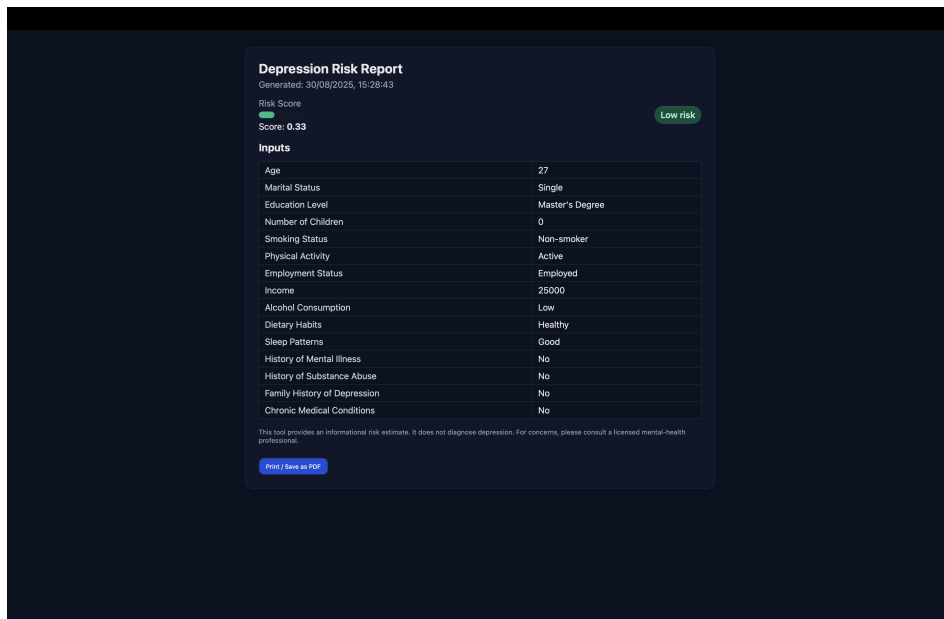


Figure D.4: PDF report

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